

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 8

1. **Hypothesis Testing:** to make a claim about a population parameter(s) or distribution(s), based on collected data.

- Hypotheses are claims about the population parameters.
- This time: one sample hypothesis test on the population mean μ .

2. **Null hypothesis and Alternative Hypothesis**

- Null hypothesis H_0 : the conservative claim
- Alternative hypothesis H_a : what the scientist wants to prove, a "radical" claim, etc
- Three cases: (i) $H_0 : \mu = \mu_0$; $H_a : \mu \neq \mu_0$; (ii) $H_0 : \mu = \mu_0$; $H_a : \mu > \mu_0$; (iii) $H_0 : \mu = \mu_0$; $H_a : \mu < \mu_0$.
- Some examples (For each of the following applications, state what a hypothesis test of interest would be, (null and alternative hypothesis))
 - (a) A new COVID-19 vaccine by Gilead, a pharmaceutical company, for the adult population in the US
 - (b) a machine learning algorithm that forecasts financial crisis (defined as a drop of the stock market index like S&P500 by more than 30%) using macroeconomic variables
 - (c) A well-known food brand (think Kraft or Cadbury) claims that there're 500g of chocolate pieces in the bag

3. **p-value**

- (a) Definition: Given a realization of the test statistic $t = T(x_1, \dots, x_n)$, the *p-value* associated with t is:
 - $\mathbb{P}(T(x_1, \dots, x_n) \geq t \mid H_0 \text{ is true})$ if $H_A : \mu > \mu_0$
 - $\mathbb{P}(T(x_1, \dots, x_n) \leq t \mid H_0 \text{ is true})$ if $H_A : \mu < \mu_0$
 - $\mathbb{P}(|T(x_1, \dots, x_n)| \geq |t| \mid H_0 \text{ is true})$ if $H_A : \mu \neq \mu_0$
- (b) intuitively, it measures the probability of observing data as extreme or more extreme than the one you've collected, assuming the null hypothesis holds.
- (c) we reject the null hypothesis if the p-value is less than α , a given user-specified significance level (typically $\alpha = 0.05$)

4. **Type-I error and Type-II error**

- (a) $\mathbb{P}(\text{Type-I error}) := \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ holds})$; False Positive Rates; A scalar
- (b) $\mathbb{P}(\text{Type-II error}) := \mathbb{P}(\text{not reject } H_0 \mid H_0 \text{ does not hold, and the true parameter is } \mu') =: \beta(\mu')$; False Negative Rates; Is a function of on True parameter μ' .
- (c) Our procedure controls the $\mathbb{P}(\text{Type-I error})$ (being less than or equal to α) and the $\mathbb{P}(\text{Type-II error})$ decreases as n increases.
 - Repeat the experiments T times, in which the null hypothesis holds, we will falsely reject H_0 among approximately $T \times \mathbb{P}(\text{Type-I error})$ experiments.
 - Repeat the experiments T times, in which the null hypothesis doesn't hold and the true parameter is μ' , we will falsely not reject H_0 among approximately $T \times \beta(\mu')$ experiments.
- (d) Trade-off between $\mathbb{P}(\text{Type-I error})$ and $\mathbb{P}(\text{Type-II error})$

- Smaller $\alpha \implies$ large $\mathbb{P}(\text{Type-II error})$.
- Extreme case $\alpha = 0$: never reject the null hypothesis, $\mathbb{P}(\text{Type-II error})$ error is 1 under regular conditions.

5. **Example** A food manufacturing facility employs an automated canning apparatus for the purpose of packaging canned food items. The predetermined standard weight for each can is 500g. A sample comprising ten randomly selected cans underwent weight measurement, yielding the following observations:

495, 510, 505, 498, 503, 492, 502, 512, 497, 506.

Assume the weight of each can, denoted by X is normally distributed $X \sim N(\mu, \sigma^2)$ with $\sigma = 6.5$. Is the machine working normally? (take significant level to be $\alpha = 0.05$)